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Seeing what plants need: AI vision and tiered automation for growers

What a photo can and cannot tell, how to predict needs before stress shows, and three fields read from orbit in 2026

HORTUS RESEARCH, LASTFOMO · DATA TO 25 OCTOBER 2026

BACKGROUND Phone cameras and AI image models promise cheap plant diagnosis, and automation promises to run a growing space on its own. Lab accuracy has not carried over to the field, and automation is usually sold as a package rather than as steps a grower can check.

AIM To set out what image analytics can and cannot tell a grower, to build a transparent analyser and a needs predictor that run in the browser, and to stage automation in tiers, tested against 2026 satellite and weather data for three real fields.

METHODS A review of evidence on image-based diagnosis outside the lab; a colour-index analyser (Excess Green with an Otsu threshold, VARI, and hue rules inside the plant outline) with an optional multimodal second opinion; a needs model built on FAO-56 Penman–Monteith reference evapotranspiration, crop coefficients, daily light integral, vapour pressure deficit and the Hutton criteria; five automation tiers mapped to controllers; and Sentinel-2 time series with NASA POWER weather for 6 × 6 km windows in Kansas (USA), Mkushi (Zambia) and West Java (Indonesia).

RESULTS Published models drop from 99.35% accuracy on lab images to 31.4% on photos taken elsewhere, and eight background pixels predict lab labels with 49% accuracy; a general multimodal model reached 56.9% zero-shot on apple leaves. In Kansas, NDVI curves separated five crop calendars; irrigated maize (46% of the window) used 735–936 mm of water against 436 mm of rain in a summer as hot as 2012, implying 520–714 mm of irrigation. Mkushi's pivots carried winter wheat through a season with 2.6 mm of rain, needing 534–644 mm. In West Java the share of rice land reaching green canopy in the dry season fell from 91% in 2025 to 62% in 2026 as July to mid-September rain fell from 304 mm to 6 mm.

CONCLUSIONS A photo is a screening tool. Predicting needs from weather and sensors is where automation earns its keep, and tiers let a grower add the next step only when the last one has paid. Evidence is strong for FAO-56 water budgeting, moderate for closed-loop fertigation, and limited for image diagnosis in the field and for full autonomy.

KEYWORDS plant phenotyping · colour indices · evapotranspiration · controlled environment agriculture · Sentinel-2 · irrigation

1. Introduction

A grower makes the same decisions every day: whether to water, how much to feed, when to vent, what to look at first. Visible symptoms come late. By the time a leaf yellows, the cause has usually been at work for days. Two technologies claim to close the gap. Image models promise to read a plant from a photo; automation promises to act on sensor data without waiting for symptoms.

Both claims need checking. Image classifiers trained on curated photos score close to perfectly on their own test sets, but those scores say little about a phone photo in a greenhouse. Automation is often sold as an integrated package, which makes it hard for a grower to see which decision is being taken, on what evidence, and whether it pays. This study asks four questions:

1. **RQ1.** How well does image-based plant diagnosis work outside the lab, and what can a photo reliably tell a grower?
2. **RQ2.** Which needs can be predicted from weather and sensor data before stress is visible, and with which methods?
3. **RQ3.** How should automation be staged so that each tier adds a decision the grower can verify?
4. **RQ4.** What do open satellite and weather data show for real fields in 2026, and how do the results connect to the tiers?

The study comes with working tools. The Hortus research pages run the analyser and the needs predictor in the browser, so a reader can test both claims on their own plants.

2. Background

2.1 IMAGE DIAGNOSIS OUTSIDE THE LAB

The PlantVillage collection released more than 50,000 expert-curated leaf images of healthy and diseased crops (Hughes & Salathé, 2015). A deep network trained on it identified 14 crops and 26 diseases with 99.35% accuracy on held-out images, but on photos from other sources, taken under different conditions, accuracy fell to 31.4%, against 2.6% for random choice (Mohanty et al., 2016). Part of the lab score is a shortcut: a model given only eight background pixels from each PlantVillage image predicted the label with 49.0% accuracy (Noyan, 2022). Singh et al. (2020) built PlantDoc, 2,598 images gathered outside the lab, because such data were missing; training on it raised classification accuracy on non-lab images by up to 31%. General multimodal models do not escape the problem. On PlantVillage apple leaves, GPT-4o reached 98.12% after fine-tuning but 56.87% zero-shot, and 69.37% zero-shot on maize (Roumeliotis et al., 2025), and those were still lab images.

2.2 WHAT COLOUR CAN MEASURE

Colour indices cannot name a disease, but they measure what is there. Excess Green ($2g - r - b$ on chromatic coordinates) separates living plants from soil and residue in sun and shade (Woebbecke et al., 1995); Otsu's (1979) method picks the threshold from the image itself. The Visible Atmospherically Resistant Index, $VARI = (G - R)/(G + R - B)$, tracks green vegetation fraction from visible bands alone (Gitelson et al., 2002). Together they give plant cover, greenness and the share of tissue that has turned yellow or brown: screening signals, not diagnoses.

2.3 PREDICTING NEEDS

Water demand is the best-established prediction in agronomy. FAO-56 computes a reference evapotranspiration (ET_0) from weather with the Penman–Monteith equation and multiplies it by a crop coefficient (Kc) for the crop and stage (Allen et al., 1998). Where only temperature is known, Hargreaves

and Samani's (1985) equation gives a cruder estimate. Crop coefficients can be scaled from satellite vegetation indices (Kamble et al., 2013). Light is budgeted as the daily light integral (DLI), the photosynthetic photons received per square metre per day (Faust & Logan, 2018); Michigan State University guidance suggests at least 12 mol m^{-2} d^{-1} for leafy greens and herbs and 15 for fruiting vegetables in controlled environments (Runkle, n.d.). Disease risk can be forecast from weather: the Hutton criteria, Great Britain's warning system for potato late blight since 2017, flag two consecutive days each with a minimum of 10 °C or more and at least six hours of relative humidity at 90% or more (Dancey et al., 2017).

2.4 AUTONOMY

In the first Autonomous Greenhouse Challenge, five teams grew cucumbers remotely for about four months, each controlling a 96 m^2 compartment with its own algorithms, against a compartment run manually by commercial growers. The organisers concluded that, in general, AI performed well in controlling a greenhouse (Hemming et al., 2019); the winning team produced more cucumbers than the growers' reference (VentureBeat, n.d.). That is one challenge, one crop and one season.

3. Methods

3.1 THE ANALYSER

The analyser runs in the browser; a photo leaves the page only if the user asks for a second opinion. Images are scaled so that the longer side is at most 640 pixels. For each pixel we compute Excess Green on chromatic coordinates, VARI, and hue, saturation and value. The plant mask is every pixel whose Excess Green exceeds an Otsu threshold (160 bins over −0.5 to 1.0, bounded between 0.03 and 0.20) and whose value exceeds 0.12. Inside the mask, pixels with hue between 40° and 75° and saturation above 0.35 count as yellowing; the rest count as healthy. Non-plant pixels count only if the plant encloses them: holes that do not reach the image edge and cover at most 2% of the frame, plus notches closed by a morphological closing of radius 0.8% of the longer side. Among those, hue 38°–75° with value above 0.35 counts as yellowing and hue 8°–38° with value below 0.75 as browning (saturation above 0.22 for both). The screening index (0–100) is the healthy share, less 0.6 times the browning share, plus up to ±5 points for mean greenness (VARI). Figure 1 shows the pipeline.

The optional second opinion sends the photo and the measured shares to Claude (Anthropic) with fixed rules: describe only what is visible, give causes as possibilities with a likelihood, prefer checks a grower can do (leaf undersides, roots, the pattern on old versus new leaves, a soil or feed test) before treatments, and never name pesticide products or rates. The answer comes back as structured fields that the report prints verbatim.

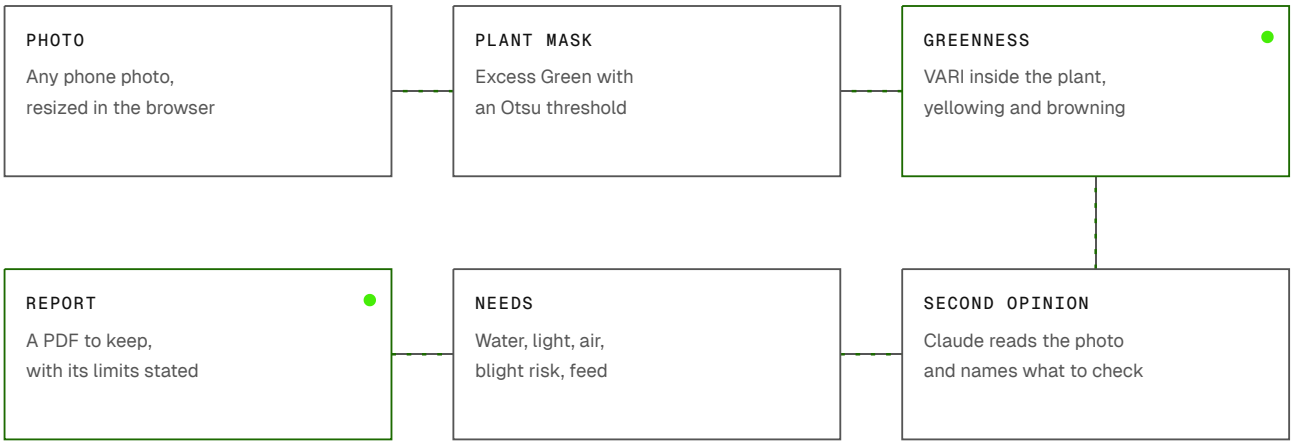


FIGURE 1. The analyser, step by step.

3.2 THE NEEDS PREDICTOR

Daily ET_0 follows FAO-56 Penman–Monteith from maximum and minimum temperature, relative humidity, wind at 2 m, solar radiation, latitude, elevation and day of year (Allen et al., 1998). Crop water use is ET_0 × Kc, with Kc from FAO-56 Table 12 for the initial, mid-season and late stages and their mean for development. Litres per plant are crop water use times the area per plant; the interval between waterings is the usable soil water (an input) divided by daily use. Under lights, DLI = PPFD × hours × 0.0036; outdoors we assume 48% of shortwave radiation is photosynthetically active at 4.57 µmol per joule, so DLI ≈ 2.19 × solar radiation in MJ m^{-2} d^{-1}. Vapour pressure deficit uses the FAO-56 vapour pressure equations. Growing degree-days use a base of 10 °C, capped at 30 °C for maize. Late blight uses the Hutton criteria; feed pH and EC are compared with a working window of pH 5.5–6.5 and with the grower's own recipe (±0.3 mS cm^{-1}); CO_2 readings are checked against the US occupational limit of 5,000 ppm as an eight-hour average (29 CFR 1910.1000). Indoor and greenhouse ET_0 use simplified radiation terms and should be calibrated.

3.3 TIERS

We staged automation by what the system senses, decides and does, and mapped each tier to Hortus controllers using their published specifications (Table 2). A tier is complete only when its decisions can be checked by the grower: a reading, a rule and an action that can be overridden.

3.4 THREE FIELDS FROM ORBIT

We read Sentinel-2 Level-2A surface reflectance (Collection 1, 10 m) from the Earth Search archive on AWS for three 6 × 6 km windows: centre-pivot farms near 38.06° N, 101.09° W in western Kansas (tile 14SLH, March–October 2026); pivots near 13.84° S, 29.24° E at Mkushi, Zambia (35LQE, January–October 2026); and lowland rice land near 6.52° S, 108.15° E in West Java (48MZT, May–October 2025 and 2026). We kept scenes with at most 50–70% cloud over the tile, masked every pixel not classed as vegetation, bare soil or water by the scene classification layer, and dropped scenes with less than 60% valid pixels in the window. Reflectance is DN × 0.0001 − 0.1. We computed NDVI (B08, B04), NDRE (B8A, B05) and NDMI (B8A, B11).

In Kansas we interpolated each pixel's NDVI to 10-day steps from 11 March to 7 October using the 34 nearly cloud-free scenes, clustered the curves with k-means (k = 5) and named the clusters from their shape. In Mkushi we classed as irrigated every pixel whose median NDVI over July and August exceeded 0.6. In West Java we took each pixel's highest NDVI across the July–September scenes of each year and counted the share above 0.5.

Daily weather came from NASA POWER on AWS: MERRA-2 temperature, dew point, wind and pressure; FLASHFlux shortwave radiation; and IMERG precipitation (MERRA-2 where IMERG was missing). For each

cropped class we set the season from the class NDVI curve (from 15 days before NDVI first reaches 0.30 to the last day at or above it), scaled Kc linearly from 0.30 at NDVI 0.15 to the mid-season FAO-56 value at NDVI 0.85 (maize 1.20, wheat 1.15, other summer crops 1.10), and computed crop water use with both Penman–Monteith and Hargreaves ET_0. Net irrigation need is the sum over 10-day blocks of crop water use minus 80% of rain, where positive. The range between the two ET_0 methods is reported as the uncertainty.

3.5 EVIDENCE LABELS

As in Study 01: STRONG EVIDENCE where results are consistent across many studies, sites or seasons; MODERATE where several studies or one solid dataset agree; LIMITED where results rest on few studies, lab conditions or a single season.

4. Results

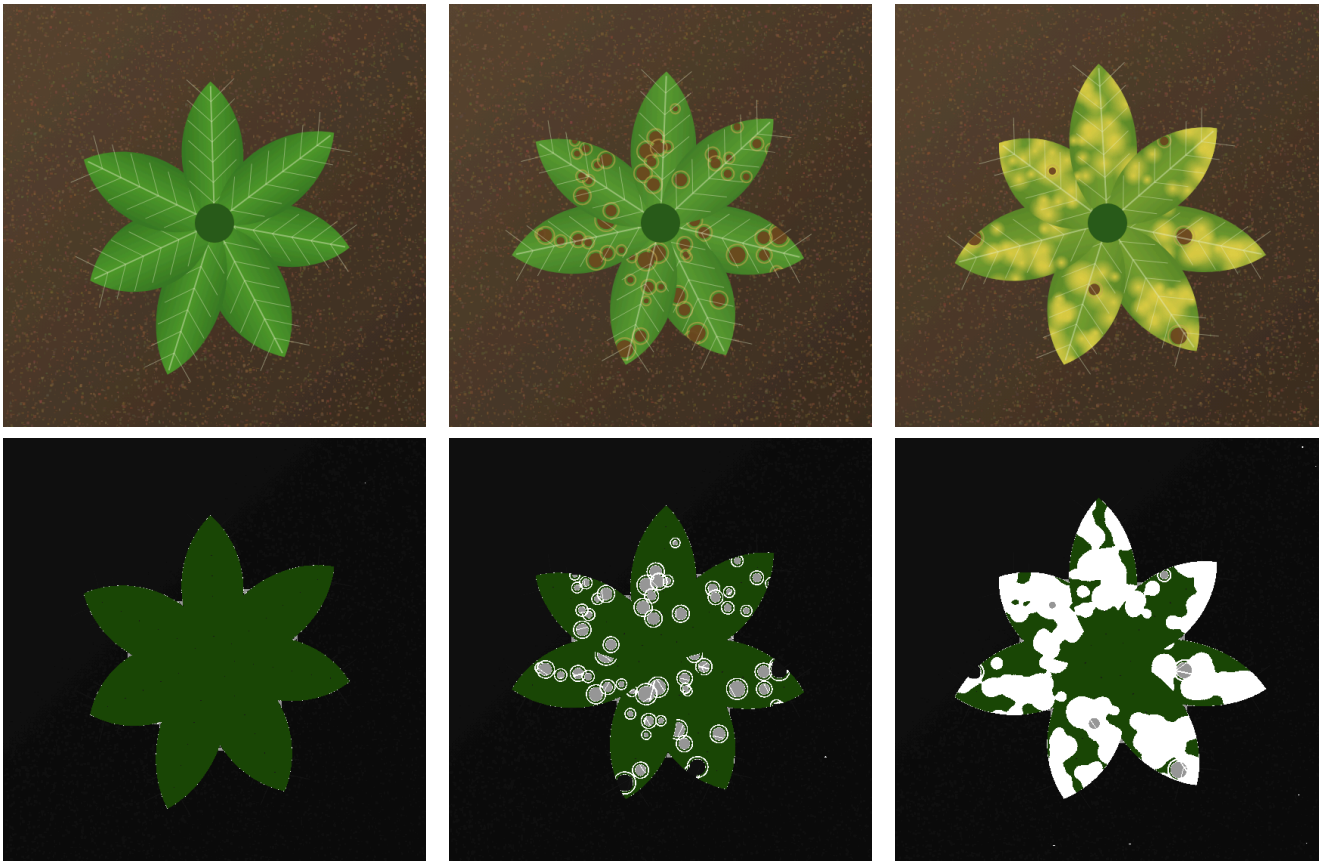
4.1 WHAT A PHOTO CAN TELL

The evidence is consistent: high accuracy on curated images, large losses on photos from elsewhere, and shortcuts that inflate lab scores (Table 1). No published result we found shows a general model diagnosing diseases reliably from arbitrary phone photos in the field. MODERATE EVIDENCE that field accuracy falls far below lab accuracy; LIMITED for any specific field accuracy.

TABLE 1. Image-based plant diagnosis: lab results against results elsewhere.

STUDY	IMAGES AND TASK	LAB RESULT	ELSEWHERE
Mohanty et al. (2016)	PlantVillage, 14 crops, 26 diseases; deep CNN	99.35% on held-out images	31.4% on photos from other sources (chance 2.6%)
Noyan (2022)	PlantVillage, background pixels only	49.0% from 8 background pixels	Shows label leakage, not plant features
Singh et al. (2020)	PlantDoc, 2,598 non-lab images, 13 species	—	Training on non-lab images: up to +31% accuracy
Roumeliotis et al. (2025)	PlantVillage apple and maize; GPT-4o and ResNet-50	98.12% (GPT-4o fine-tuned, apple)	Zero-shot GPT-4o: 56.87% apple, 69.37% maize

On three synthetic test images the analyser behaves as designed (Figure 2). The healthy plant scores 99.6% healthy tissue and a screening index of 100; the plant with leaf spots scores 82.2% healthy, 9.0% yellowing (the halos) and 8.8% browning (the spots), index 81; the plant with yellowing patches scores 46.0% healthy and 52.6% yellowing, index 45. Before the enclosed-region rule, soil at the leaf edges was read as browning and the healthy plant scored only 77% healthy; restricting off-mask tissue to regions the plant encloses removed that error. We have not yet validated the analyser on labelled field photos. LIMITED EVIDENCE



Healthy: 99.6% healthy, 0.2% yellowing, 0.2% browning; index 100

Leaf spots: 82.2% healthy, 9.0% yellowing, 8.8% browning; index 81

Yellowing: 46.0% healthy, 52.6% yellowing, 1.4% browning; index 45

FIGURE 2. The analyser on three synthetic images: photo (top) and stress map (bottom). Stress map: yellowing white, browning grey, healthy green dimmed.

4.2 PREDICTING NEEDS

For irrigated maize at mid-season under the June–August 2026 means for the Kansas window (maximum 36.5 °C, minimum 21.4 °C, 47% relative humidity, wind 3.5 m s^{-1}, 26.5 MJ m^{-2} d^{-1}), the model gives ET_0 8.1 mm d^{-1} and crop water use 9.7 mm d^{-1}, or 1.26 L per plant per day at 0.13 m^2 per plant; the vapour pressure deficit is 2.3 kPa and the daily maximum crosses the 35 °C heat flag. For a mid-season greenhouse tomato under the example readings (27/17 °C, 70% relative humidity, 12 MJ m^{-2} d^{-1} inside), use is 2.9 mm d^{-1}, 1.4 L per plant at 0.5 m^2, light 26 mol m^{-2} d^{-1} (above the 15 minimum), and one of the two Hutton days is met. These are estimates whose method is mature; the inputs, especially wind and radiation inside a greenhouse, decide their accuracy. STRONG EVIDENCE for the method; MODERATE for greenhouse and indoor inputs

4.3 FIVE TIERS

The tiers add one kind of decision at a time (Table 2; Figure 3). Tier 0 needs only a phone and screens for stress. Tier 1 senses the root zone and the air and acts on thresholds. Tier 2 closes control loops on feed, pH, climate and CO_2. Tier 3 predicts: it moves set points with the weather forecast and the crop's stage, and warns before stress shows. Tier 4 optimises: it trades yield against energy, water and CO_2, with people auditing the system. Each tier keeps the one below working on its own, so a network or cloud failure degrades the system by one step instead of stopping it.

TABLE 2. Tiers: what each senses, decides and does, the hardware that runs it, and the strength of evidence.

TIER	SENSES	DECIDES	DOES	RUNS ON	EVIDENCE
0 · Look	Photo: colour indices (ExG, VARI); Your notes	What to inspect first; When to ask an expert	You water, feed and treat by hand	This page	Limited
1 · Sense	Root-zone moisture; Air temperature and humidity	Water when the roots ask; Heat or cool on two channels	One pump on a moisture trigger; Heater and cooler sockets	HortusRoom S: waters from a moisture reading, 10 s to 60 min, pause 1 min to 10 h, pump up to 1,500 W; HortusRoom A: two channels up to 1,500 W each, –40 to +80 °C	Moderate
2 · Control	EC and pH of the feed; Solution and air temperature; CO_2	Feed to the set EC, then correct pH; Raise CO_2 while the light is on	Peristaltic dosing of up to three nutrients; pH+ and pH– pumps; CO_2 regulator or fan	HortusPilot N: EC 0.1–5.0 mS, pH 3.0–14.0, tanks 10–2,000 L; HortusPilot F: 15 sensors, 26 alarms, RS-485 Modbus; HortusPilot pH: PI control, pH 3.0–12.0; HortusRoom C: CO_2 300–5,000 ppm, threshold 400–2,000 ppm	Moderate
3 · Predict	Weather: temperature, humidity, wind, radiation; Satellite canopy (NDVI, NDRE, NDMI); Logs from tiers 1–2	Water demand from $ET_0 \times Kc$; Light and heat budgets; Blight periods ahead of time	Set points move with the forecast; Alerts before stress shows	Tier 2 hardware plus a data link (Wi-Fi or RS-485)	Moderate
4 · Autonomy	All of the above, continuously	Trade yield against energy, water and CO_2 cost; Plan days ahead	The system runs the room; people audit it	A climate computer on top of tiers 1–3	Limited

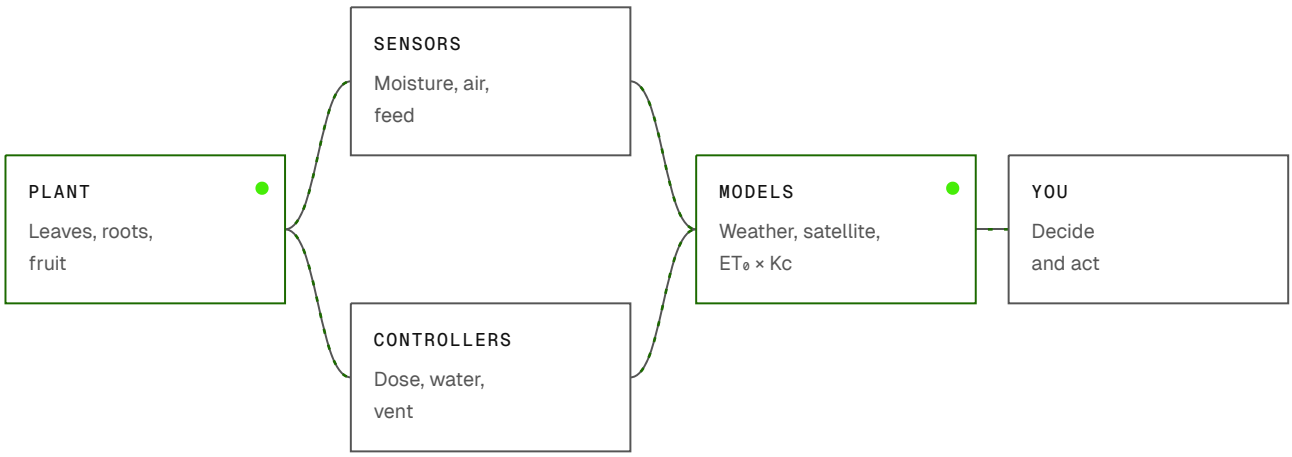


FIGURE 3. What tier 3 connects: plant, sensors, controllers, models and the grower.

4.4 KANSAS: FIVE CROP CALENDARS AND A HOT SUMMER

NDVI curves separated five crop calendars in the 36 km² Kansas window (Figures 4 and 5): irrigated maize, peaking at NDVI 0.87 in late July (45.7% of the window); winter wheat followed by a second summer crop (16.3%); winter wheat followed by fallow (8.8%); a late-sown summer crop peaking in mid-August (12.0%); and dryland, fallow and field corners (17.1%). The summer was hot: the June–August mean daily maximum was 36.5 °C, 2.9 °C above the 1991–2020 normal and level with 2012 (36.4 °C); 75 days from May to September reached 35 °C.

From 27 May to 1 October the irrigated maize used an estimated 735 mm (Hargreaves) to 936 mm (Penman–Monteith) against 436 mm of rain, for a net irrigation need of 520–714 mm (Figure 6). The Penman–Monteith total sits high; reanalysis wind and humidity on a half-degree grid may overstate

demand at field scale, and the budget carries no soil-water storage from one 10-day block to the next. The double-cropped fields needed more over their two crops (761–1,038 mm). MODERATE EVIDENCE

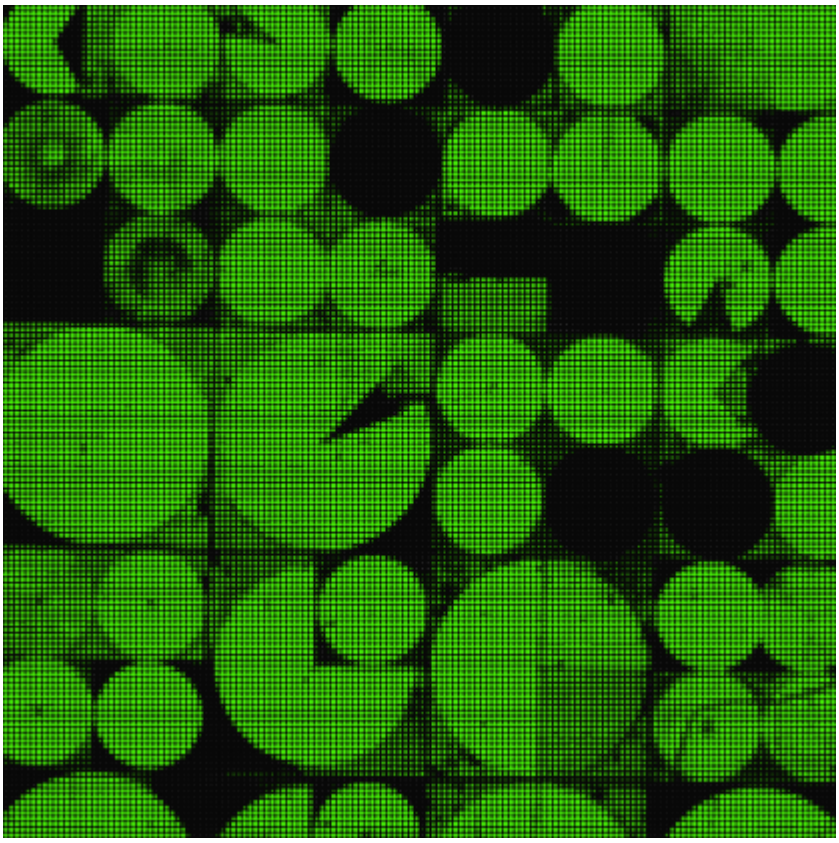


FIGURE 4. Kansas window at peak canopy (NDVI, 30 July 2026, 6 × 6 km). Bright dots are high NDVI; grey dots are bare or senescent ground.

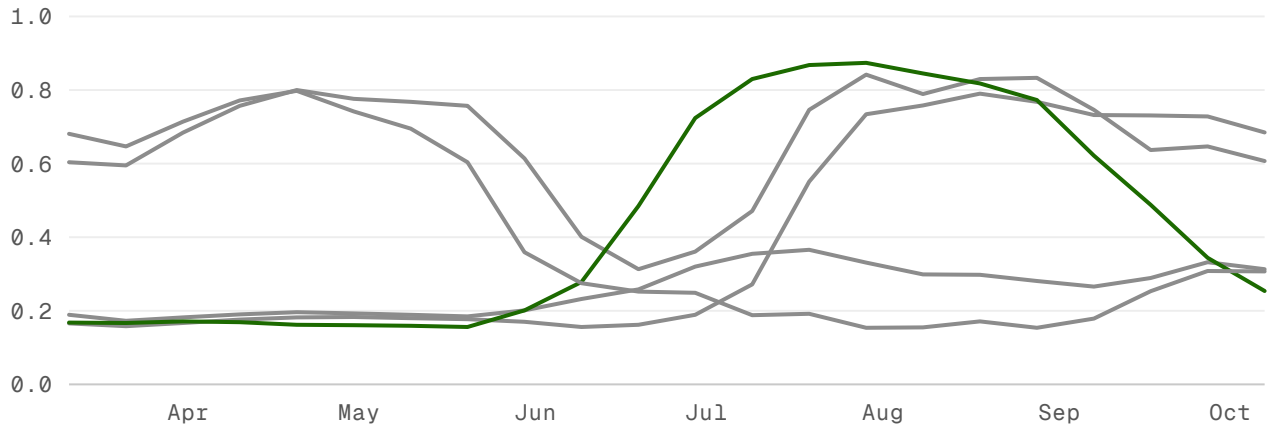


FIGURE 5. NDVI curves of the five classes, 10-day steps, 2026. Highlighted: irrigated maize.

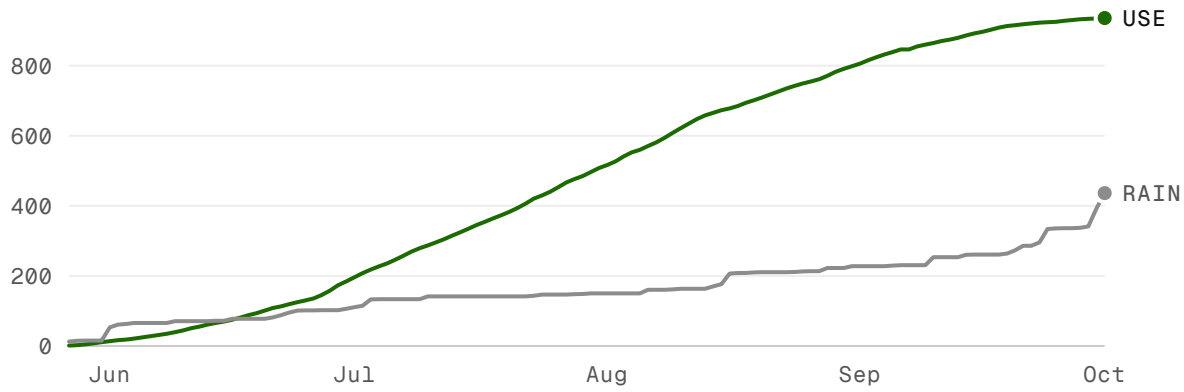


FIGURE 6. Irrigated maize: cumulative crop water use (Penman–Monteith) against rain, 2026.

4.5 MKUSHI: WATER ONLY FROM THE PIVOT

At Mkushi, pivots covered 25.6% of the window and stood out against the land around them through the dry season: pivot NDVI peaked at 0.87, while the rest of the window dried down to 0.22 (Figure 7). From 20 April to 24 September the winter wheat used an estimated 536–646 mm while 2.6 mm of rain fell, so practically all of it came from irrigation (534–644 mm). Daily demand peaked at 6.6 mm. MODERATE EVIDENCE

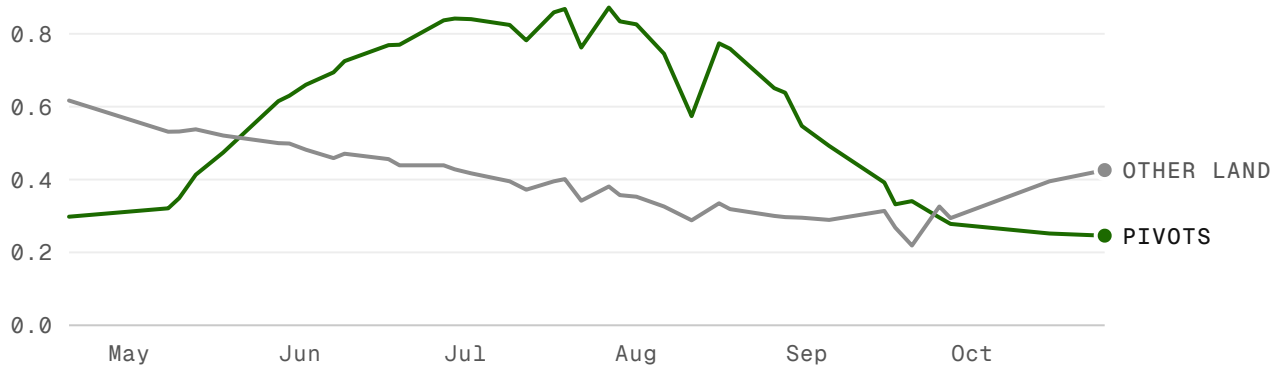


FIGURE 7. NDVI of the irrigated pivots and of everything else in the Mkushi window, 2026.

4.6 WEST JAVA: THE DROUGHT, SEEN FROM ORBIT

In the West Java window, 91.5% of the land reached an NDVI of 0.5 at some point in July–September 2025, against 62.4% in 2026; the median of each pixel's seasonal maximum fell from 0.76 to 0.58 (Figure 8). Rain from 1 July to 14 September fell from 304 mm in 2025 to 6 mm in 2026. The images show large bare and dry patches where rice stood a year earlier. This is the El Niño drought measured in Study 01 (Java's July–September rain −85%), seen field by field. MODERATE EVIDENCE (two seasons, one window)



FIGURE 8. West Java, true colour, 10 August 2025 (left) and 13 August 2026 (right). Contains modified Copernicus Sentinel data 2025–2026.

TABLE 3. The three windows: scenes used and main results.

WINDOW	SCENES USED	MAIN RESULT	WATER OR RAIN
Kansas, USA · 14SLH · 2026	39 (34 near cloud-free)	Five crop calendars; irrigated maize 46% of the window	Maize use 735–936 mm; rain 436 mm; irrigation 520–714 mm
Mkushi, Zambia · 35LQE · 2026	47 (38 near cloud-free)	Pivots 26% of the window	Wheat use 536–646 mm; rain 2.6 mm
West Java, Indonesia · 48MZT · 2025–26	53 (14 + 21 in Jul–Sep)	Green in the dry season: 91% (2025), 62% (2026)	Rain 1 Jul–14 Sep: 304 mm (2025), 6 mm (2026)

5. Discussion

RQ1. A photo reliably tells a grower where to look: how much of the plant is green, how much has yellowed or browned, and whether that share is growing between photos of the same plant. It does not reliably tell why. Yellowing can come from nitrogen shortage, waterlogging, cold, age or a virus; the colour is the same. The published gap between lab and field accuracy, and the shortcut that background pixels offer, argue for a screening tool with stated limits rather than a diagnosis button. A multimodal model is useful as a second opinion that names checks, not as an oracle.

RQ2. Water, light, heat and some disease risks can be predicted from weather before a plant shows stress, with methods decades old and well tested. Kansas in 2026 shows the stakes: at 9–10 mm a day in July, a maize field draws down 60–70 mm of soil water a week, and irrigation has to keep pace through flowering. The prediction is only as good as its inputs, which is why tier 3 sits on top of the sensors of tiers 1 and 2: on-site temperature, humidity and soil moisture correct the reanalysis.

RQ3. Tiers make the economics legible. Moisture-triggered watering (tier 1) replaces watering by the calendar with watering by need; closed-loop feed and pH control (tier 2) is widely used in hydroponics, and its weak point is the probes, which drift and need regular calibration; prediction (tier 3) moves the set points before the weather turns; autonomy (tier 4) remains a research frontier with one strong demonstration. A grower can stop at any tier and still have a working system.

RQ4. Open satellite and weather data already show what each field needed this year, at 10 m, within days. The same pipeline that separates crop calendars in Kansas and finds dry rice land in Java can set Kc and irrigation targets for a tier-3 controller, or flag a field whose canopy falls behind its class.

LIMITATIONS

- **No field validation of the analyser.** The thresholds are set from the literature and tested on synthetic images only.
- **Water budgets** use reanalysis weather, a linear NDVI–Kc scaling with assumed end points, a fixed 80% rain efficiency and no soil storage; the two ET_0 methods differ by about a quarter.
- **Crop classes** are unsupervised and named from curve shape; we did not check them against field records.
- **Thresholds for pH, EC and VPD** are practitioner working ranges, not crop-specific targets.
- **Conflict of interest.** Hortus sells the tier 1–2 controllers named here; the tier logic does not depend on them.

6. Recommendations

- **Home and small growers:** start at tier 0–1. Photograph the same plant from the same spot weekly and watch the trend; add a moisture-triggered pump before anything else. MODERATE
- **Greenhouse and hydroponic growers:** close the feed and pH loops (tier 2) and calibrate probes on a schedule; drift is the default. MODERATE
- **Field growers with irrigation:** budget water with $ET_0 \times Kc$ from local weather and check it against a soil probe; in a summer like Kansas 2026, plan for 9–10 mm a day at peak. STRONG
- **Everyone:** treat an image diagnosis as a reason to look closer, never as a reason to spray. STRONG

7. Conclusion

A photo is the cheapest sensor on a farm and the easiest to over-trust. Used as a screen, it tells a grower where to look; combined with weather, sensors and models, it becomes part of a system that predicts what a plant needs before the plant shows it. Built in tiers, that system can grow with the grower, one checked decision at a time.

Data availability. Copernicus Sentinel-2 Level-2A Collection 1 via Earth Search on AWS (contains modified Copernicus Sentinel data 2025–2026); NASA POWER daily MERRA-2, FLASHFlux and IMERG archives on AWS. Data were read on 26 October 2026.

Code availability. The processing scripts are published with the Hortus research pages (code/study-02); the analyser and needs model run in the pages themselves.

Use of AI tools. Data retrieval, analysis code, the in-browser tools, figures and the first draft of this text were produced with Claude (Anthropic), an AI model, working for the Hortus team. The analyser's optional second opinion also uses Claude, at the user's request. Every reference was checked against an online record on 26 October 2026. The Hortus team is responsible for the content. AI tools are not authors.

Funding and conflicts of interest. No external funding. Hortus designs and sells the HortusRoom and HortusPilot controllers named in Table 2.

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